DIGITAL SYSTEM DESIGN

FPGA StopWatch & Clock

Verilog-HDL & Basys3 Board Implementation

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프로젝트 개요 (Overview)

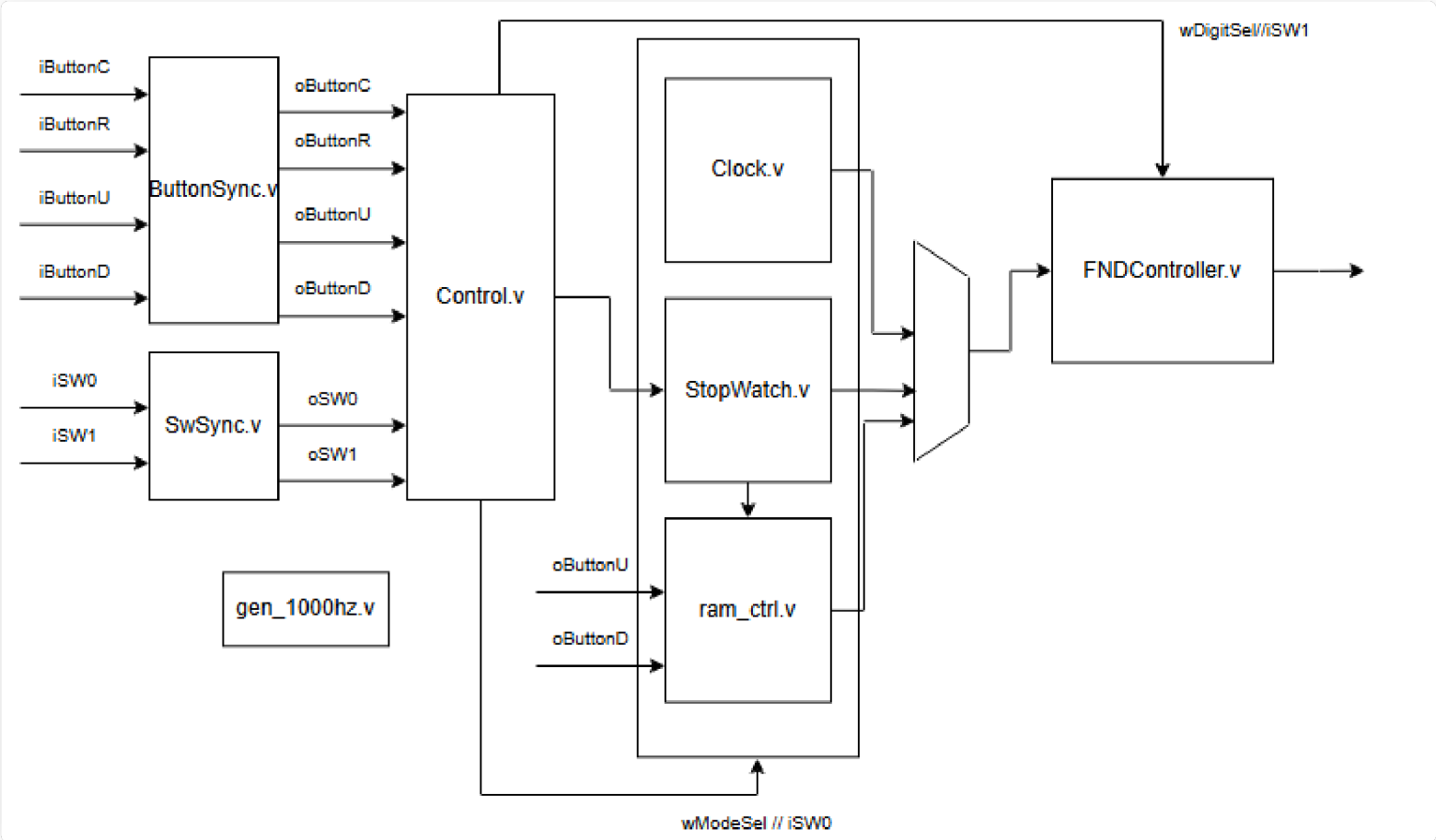
🎯 프로젝트 목표

- **Verilog-HDL 설계:** 순차/조합 논리회로를 이용한 디지털 시계 및 스톱워치 구현
- **FPGA 검증:** Basys3 (Artix-7) 보드 상에서 실제 하드웨어 동작 확인
- **Full Flow:** RTL 설계부터 Synthesis, Implementation, Bitstream 생성까지 전 과정 수행

⚙️ 주요 사양 (Specs)

- **System Clock:** 100 MHz Input
- **Display:** 4-Digit 7-Segment Display (FND)
- **Feature 1:** Digital Clock (시/분/초 설정 가능)
- **Feature 2:** StopWatch (10ms 정밀도, Run/Stop)
- **Feature 3:** Lap Time Recording (BRAM 활용)

설계 전 예상 구조 (Initial Design)



초기 블록 다이어그램 (Control Unit 중)

설계 컨셉 - 타이밍 및 제어 (Timing & Control)

Core Logic Strategy

1. Button Input

Shift Register 기반 Debouncing 회로를 통해 채터링(Noise)을 제거하고, 안정적인 Edge 신호를 추출합니다.

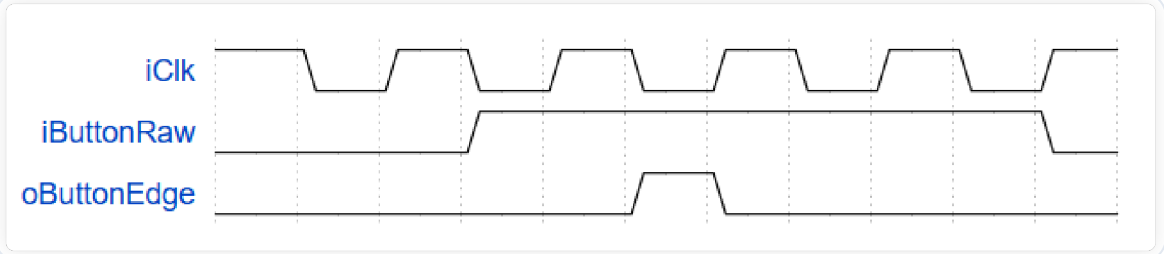
2. Clock System

100MHz 시스템 클럭을 분주(Prescaling)하여 10ms(Stopwatch), 1s(Clock) 단위 신호를 생성합니다.

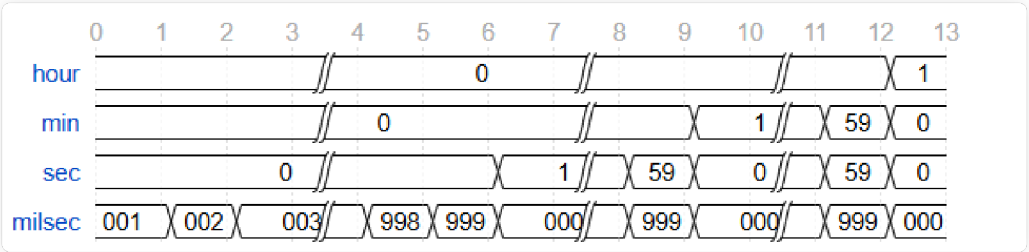
3. FSM Control

IDLE ↔ RUN ↔ STOP 상태 천이(State Transition)를 통해 스톱워치 동작을 제어합니다.

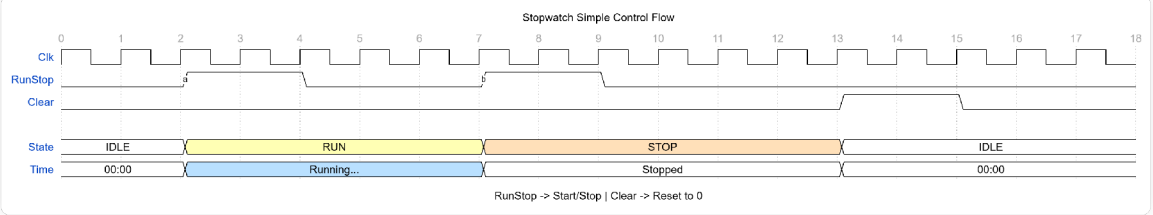
Timing — Button Debouncing



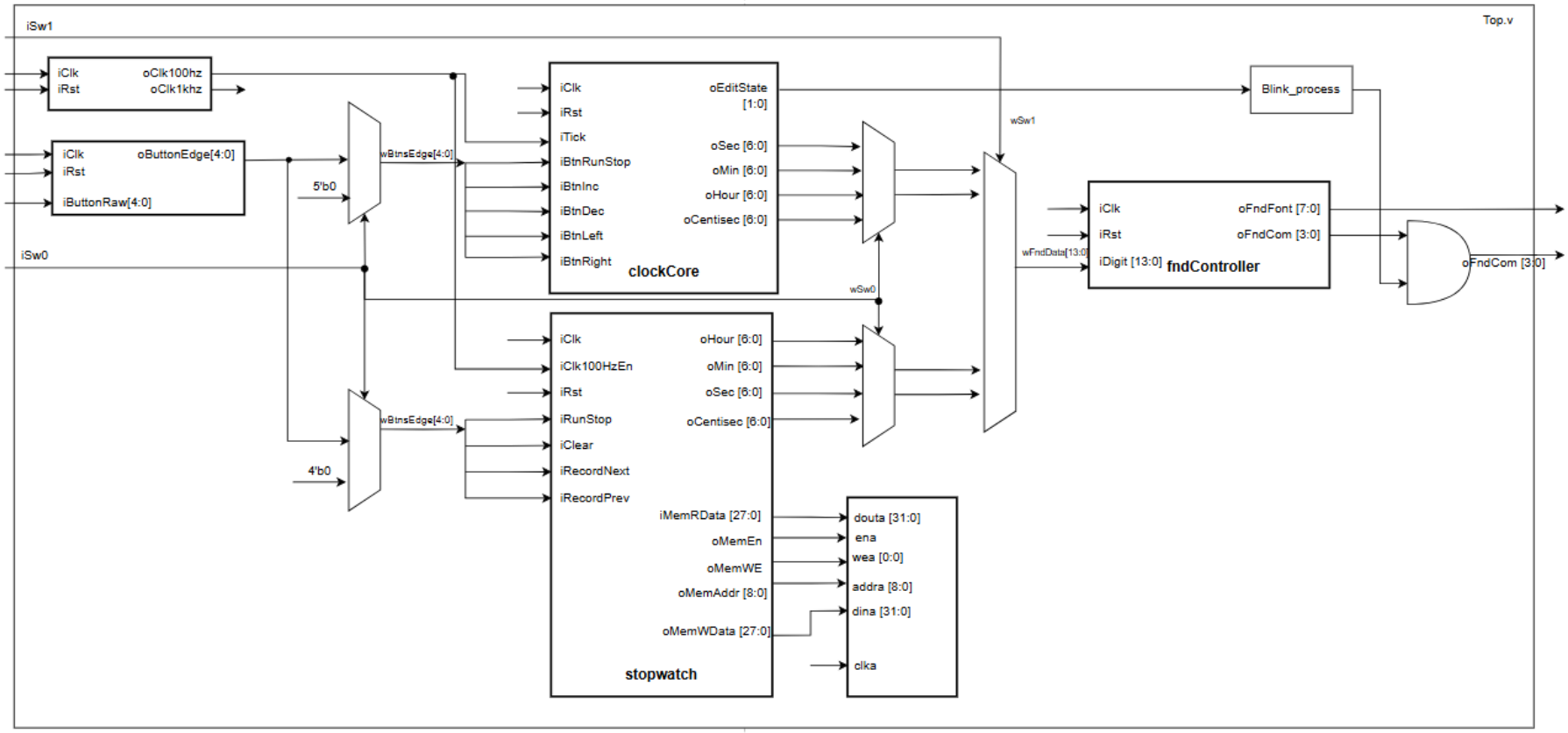
Clock — Clock Counting



FSM — Control Flow



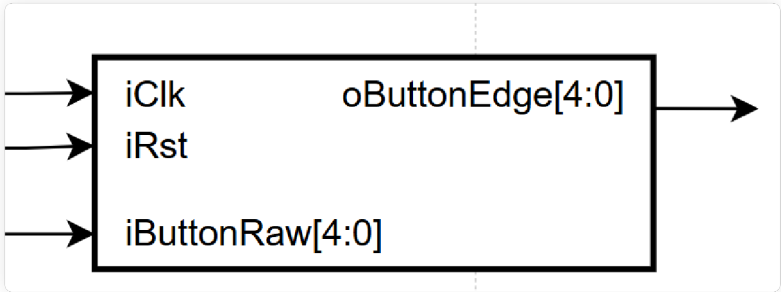
최종 설계 구조 (Final Architecture)



최종 설계 구조 (모듈화 및 BRAM IP 적용)

MODULE DETAIL

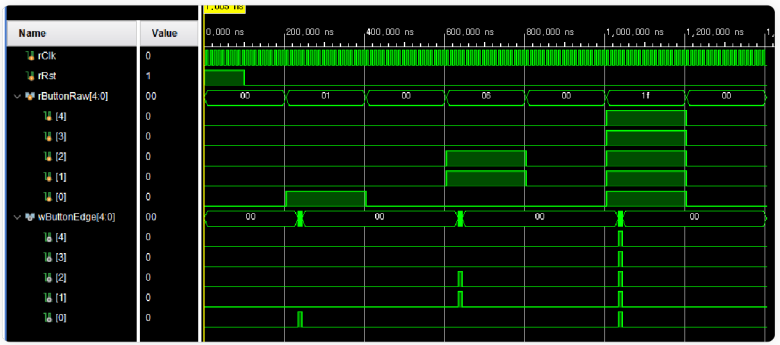
입력 동기화 (Button Sync)



Key Logic

- **Debouncing:** Shift Register를 이용해 기계적 노이즈 제거
- **Synchronization:** 100MHz 시스템 클럭에 입력 동기화
- **Edge Detect:** Posedge 검출로 1 Cycle Pulse 생성

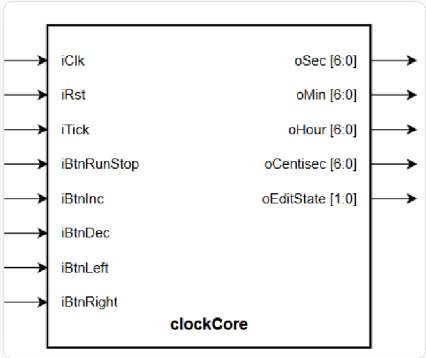
Simulation Result

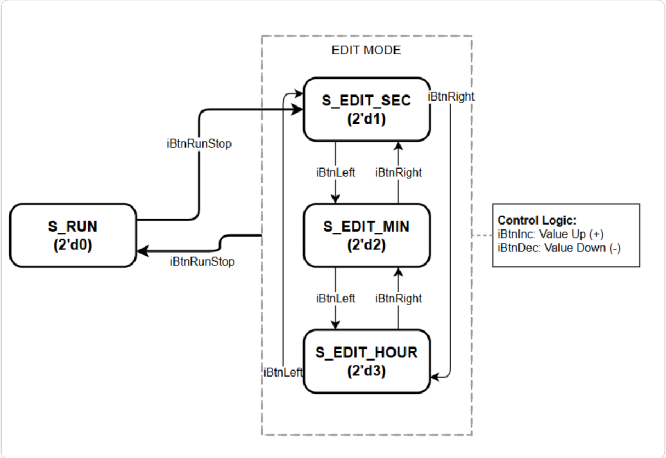


Debouncing 동작 검증 (안정적인 Edge 검출)

MODULE DETAIL

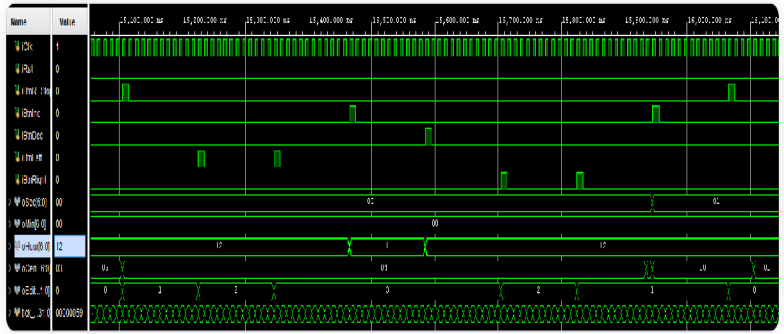
시계 코어 (Clock Core)





Module Symbol & Edit Mode FSM

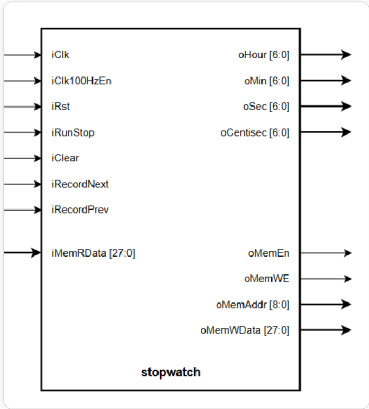
Simulation Result

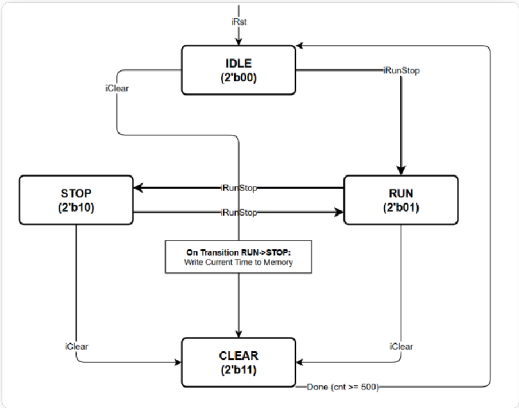


시간 카운팅 및 설정 모드 동작 검증

MODULE DETAIL

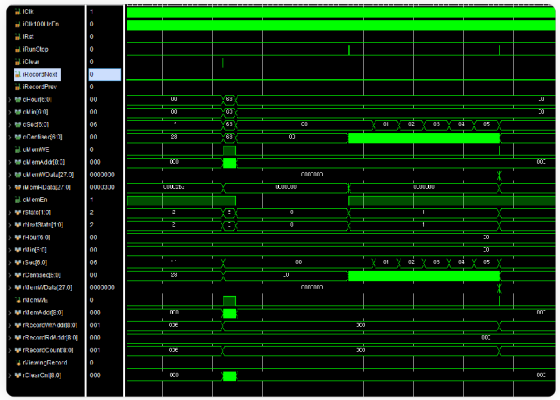
스톱워치 코어 (StopWatch Core)





Module Symbol & Control FSM

Simulation Result



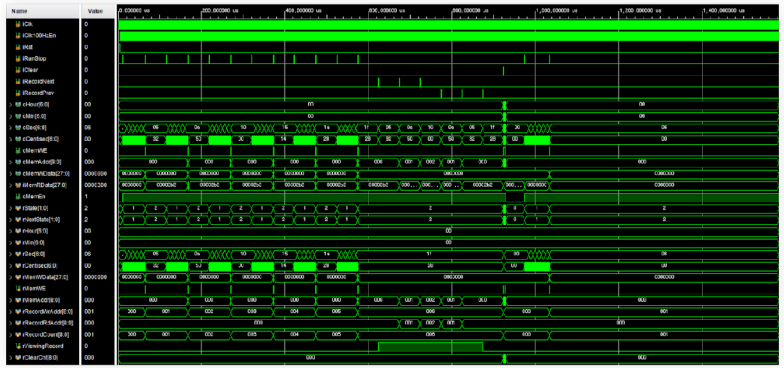
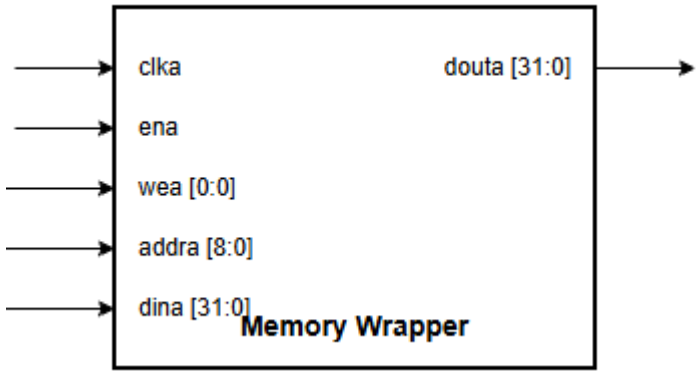


Figure 4. Run/Stop 동작 및 Memory Write 검증

MODULE DETAIL

Bram Wrapper(SinglePortRam_500x32)



FPGA BRAM

- **Single-Port BRAM:** 단일 포트 구조로 읽기/쓰기 동작
- **Control Logic:** Enable / Write Enable 기반 메모리 제어
- **Data Flow:** Lap Time 데이터 저장 및 출력

Vivado Summary

Information

Memory Type: Single Port Memory

Block RAM resource(s) (18K BRAMs): 1

Block RAM resource(s) (36K BRAMs): 0

Total Port A Read Latency : 2 Clock Cycle(s)

Address Width A: 9

CHAPTER 5. RESULT

구현 결과 (Implementation)

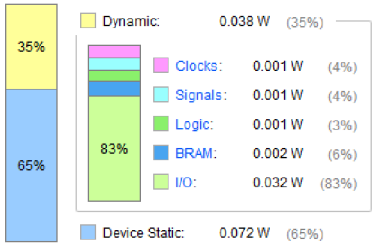
⚡ Power Analysis

Summary

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power:	**0.11 W**
Design Power Budget:	**Not Specified**
Process:	typical
Power Budget Margin:	**N/A**
Junction Temperature:	**25.6°C**
Thermal Margin:	59.4°C (11.8 W)
Ambient Temperature:	25.0 °C
Effective ϑJA:	5.0°C/W
Power supplied to off-chip devices:	0 W
Confidence level:	Low

On-Chip Power



Dynamic:	0.038 W	(35%)
Clocks:	0.001 W	(4%)
Signals:	0.001 W	(4%)
Logic:	0.001 W	(3%)
BRAM:	0.002 W	(6%)
I/O:	0.032 W	(83%)
Device Static:	0.072 W	(65%)

Total On-Chip Power: **0.11 W**

⏱ Timing Summary

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	3.970 ns	Worst Hold Slack (WHS):	0.187 ns	Worst Pulse Width Slack (WPWS):	4.020 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	288	Total Number of Endpoints:	288	Total Number of Endpoints:	175

All user specified timing constraints are met.

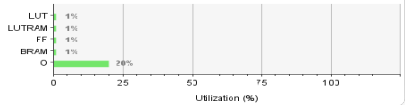
WNS: 3.970ns (Timing Met)

CHAPTER 6. CONCLUSION

리소스 사용량 및 결론

Summary

Resource	Utilization	Available	Utilization %
LUT	282	20800	1.36
LUTRAM	4	9600	0.04
FF	169	41600	0.41
BRAM	0.50	50	1.00
IO	21	106	19.81



Name	Slice LUTs (20800)	Slice Registers (41600)	F7 Muxes (16300)	Slice (8150)	LUT as Logic (20800)	LUT as Memory (9600)	Block RAM Tile (50)	Bonded IOB (106)	BUFGCTRL (32)
Top	282	169	6	95	278	4	0.5	21	1
U_clockCore (clockCore)	53	28	0	22	53	0	0	0	0
U_fndController (fndController)	4	2	0	2	4	0	0	0	0
U_gen_clk (gen_clk)	46	39	0	23	46	0	0	0	0
U_MemWrapper (stopwatch_mem_wrapper)	1	0	0	1	1	0	0.5	0	0
U_stopwatch (stopwatch)	152	56	6	47	152	0	0	0	0
U_SyncButtons (button_sync)	25	18	0	18	21	4	0	0	0

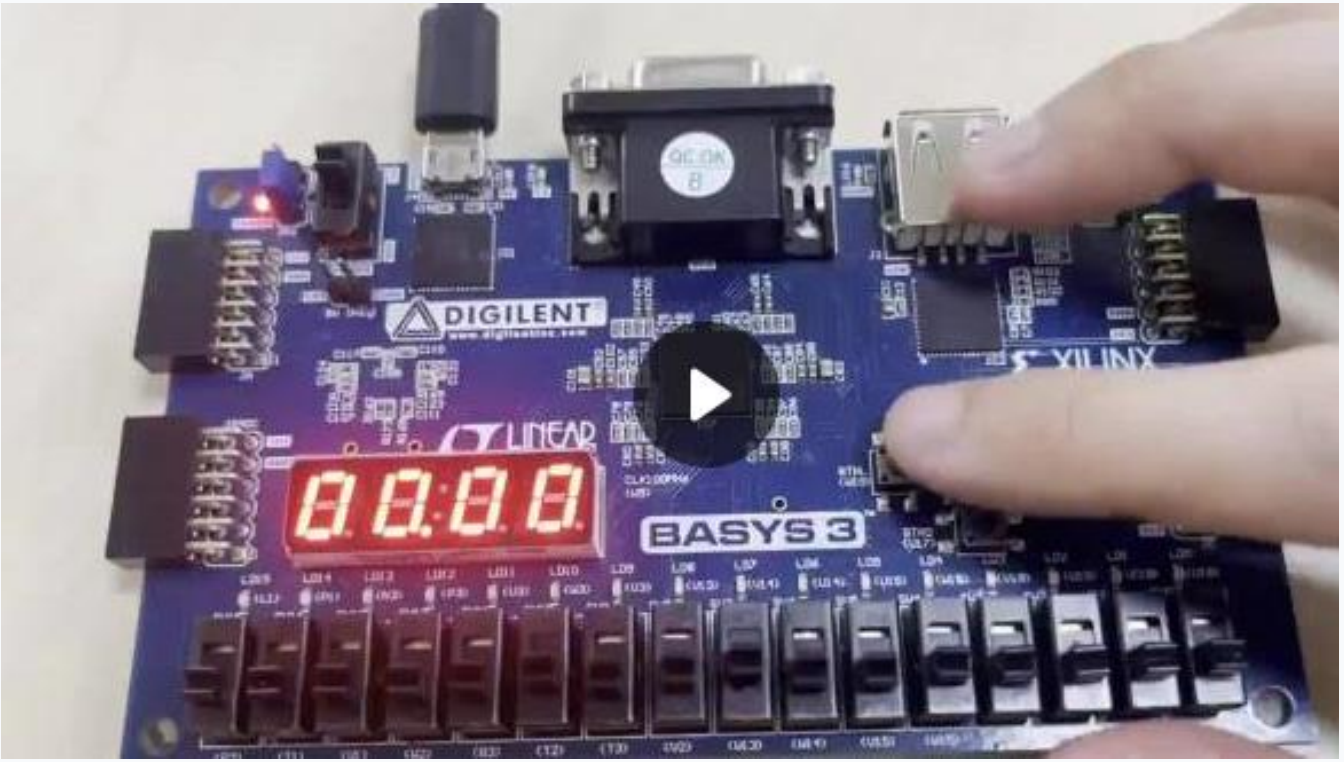
Resource Utilization (매우 효율적인 자원 사용)

📝 Project Review

- **제어구조:** FSM 기반 상태 전이로 Clock / Stopwatch 동작 흐름을 안정적으로 구성
- **기능구현 :** BRAM 기반 Lap Time 저장 구조로 기록 관리 기능 구현
- **최적화:** FPGA 자원 사용 최소화 및 Timing Constraint 충족

CHAPTER 7. DEMONSTRATION

시연 영상 (Demo Video)



Thank You

Q & A